

The Problem, And Our Approach

THE PROBLEM

Choosing bathroom fixtures means juggling room dimensions, a fixed budget, personal style, and whether things will even physically fit — today customers guess, or wait on a designer.

OUR APPROACH

- A conversational AI assistant gathers real constraints — room size, budget, style, shower size — through dialogue.
- A grounded RAG + deterministic engine picks real, priced Kohler products — never hallucinated.
- A live, interactive 2D floor plan renders the recommendation instantly — drag, rotate, delete fixtures on the fly.

WHAT MAKES IT WORK

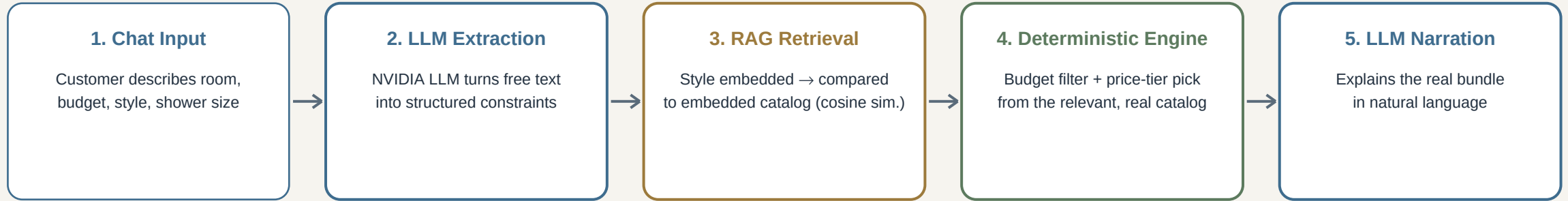
Real catalog data — 42 SKUs cross-verified against Kohler's official India price book by exact product code.

Explainable by design — the LLM never invents a price or product name; it only interprets language and explains results a deterministic engine already computed.

Room-size-aware — bathtub is automatically excluded on small rooms, offered as a choice on larger ones.

Grounded RAG — style retrieval works from real semantic meaning, not just keyword matches.

How A Message Becomes A Priced, Placed Bundle



Split-Screen Output

Left panel: full chat transcript with the recommended bundle and real prices, kept as a permanent record.

Right panel: live 2D floor plan built from real product dimensions — drag to reposition, rotate 90°, switch between 3 wall-arrangement templates, or delete a fixture (bill total updates immediately).

Only toilet, vanity, tub, shower, and faucet get floor/counter positions — concealed items (tank, mirror) are listed, never invented a footprint for.

Built End-To-End, Nothing Mocked

Frontend	Backend	AI / NVIDIA NIM	Data
React 18 + Vite Split-screen layout: chat panel + live interactive 2D SVG grid Pointer-event dragging, rotation, layout templates	Node.js + Express Stateful chat sessions (per-conversation slot filling) Deterministic recommend- ation engine (engine.js)	Chat LLM: constraint extraction + narration Embedding model: catalog + query embeddings for RAG OpenAI-compatible API (integrate.api.nvidia.com)	42 SKUs, 7 categories Sourced from Kohler's real India price book + retail spec-sheet PDFs Every price cross-verified by exact product code

Why This Isn't Just A Chatbot With A Catalog

DIFFERENTIATORS

- **Grounded RAG, not vibes**

Retrieval ranks real catalog items by semantic similarity to the customer's own words — finds “spa-like” products without needing the word “zen”.

- **Explainable, not a black box**

Every price, dimension, and product name traces back to deterministic code and verified catalog data — the LLM only interprets and narrates.

- **Physically-aware layout**

Bathtub eligibility, shower sizing (user-specified, in feet), and fixture placement all respect real room dimensions and clearances.

- **Genuinely interactive, not static**

Drag, rotate, delete, and switch layout templates — the bill updates live as the customer edits their own plan.

WHAT'S HONESTLY SIMPLIFIED

Session state is in-memory (demo-scale; swap for a real store in production).

Shower enclosure floor space uses the customer's stated size, or a generic footprint if unstated — catalog data itself has no enclosure dimensions, only fixture specs.

Fixed rectangular room only — not L-shaped layouts.

Overlap detection is a visual warning, not a hard block — prioritizes free rearrangement over rigid constraints.

Built transparently — every simplification is documented, not hidden.